Yueran (Hannah) Sun

Master's Student in Data Science | University of Washington

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Education

2025-Present **University of Washington** — **M.S. in Data Science** (Expected Mar 2027) **Seattle, WA**
- **GPA: 4.0/4.0**

2021-2025 **University of Michigan** — **B.S. in Data Science**, Minor in Mathematics **Ann Arbor, MI**
- **GPA: 3.71/4.0** (University Honors)

Research Interests

Human–Computer Interaction, Human-Centered AI, Information Seeking, Accessibility, AI & Society, AI Safety & Alignment

Research Experience

Present – Sep 2026 **Research Fellow**, Supervised Program for Alignment Research **Berkeley, CA (Remote)**
Mentor: Emilio Barkett
- Investigating how social influence emerges in LLM collaborative problem solving by analyzing when and why models revise their judgments.
- Developing counterfactual and mechanistic analyses to identify causal drivers of judgment revision, including social commitment and psychological reactance.

Present – Sep 2025 **Research Collaborator**, UW Language Accessibility Research Lab **Seattle, WA**
PI: Prof. Lucy Lu Wang
- Conducting qualitative annotation of chronic illness discussions on Reddit to characterize accessibility barriers and solutions related to symptom management.
- Evaluated AI-assisted scholarly reading interfaces in a classroom study comparing baseline reading, on-demand augmentation, and elaborative interrogation, assessing immediate and delayed comprehension, cognitive load, and reader preferences.
- Examined gaps between automated readability metrics, AI-generated adaptations, and the plain-language translation priorities of disability advocacy organizations.

Present – Sep 2025 **Research Collaborator**, UW Lab for Computing Cultural Heritage **Seattle, WA**
PI: Prof. Benjamin Charles Germain Lee
- Engaging in an independent study on multimodal linking of archival photographs and oral-history transcripts to support interactive exploration of derived entities and relationships.
- Developed a multimodal retrieval pipeline using CLIP and DBSCAN to identify, cluster, and semantically characterize reprinted photographs across 1.5M historical newspaper images.

Jul 2026 – Jun 2026 **Graduate Research Assistant**, Humanities Data Science Summer Institute **Seattle, WA**
American Institute of Physics Team, *PI: Prof. Benjamin Charles Germain Lee*
- Mentored four undergraduate researchers to improve discovery across multimodal archives, transforming 1,700+ oral histories into knowledge graphs and linking unnamed collaborators to external knowledge bases.

May 2024 – Nov 2023 **Undergraduate Research Assistant**, MobiDrop Co., Ltd. **Shanghai, China (Remote)**
Mentor: Dr. Guantao Zheng
- Pretrained a Transformer-based model for single-cell gene expression on 300,000 human blood cells, achieving a 6% reduction in validation error on masked gene expression prediction.

Aug 2023 – Jul 2023 **Undergraduate Research Assistant**, Macao Polytechnic University **Macao SAR**
Mentor: Dr. Patrick Pang
- Adapted BERTopic for topic modeling of Mandarin Chinese patient–physician discussions from an online healthcare consultation platform.

Industry Experience

Aug 2025 – Jun 2025 | Software Engineer Intern, Ternary | Seattle, WA (Remote)
Supervisor: Kyle Rattet
- Enhanced the existing cloud cost prediction system with time-series forecasting to improve cost projections and deliver interactive forecasts through client dashboards.

Dec 2024 – Jan 2024 | Student Developer, University of Michigan Multidisciplinary Design Program | Ann Arbor, MI
ProQuest Team, *Mentors: Prof. Sindhu Kutty, John Dillon, Dan Hepp*
- Developed an MLOps pipeline using large language models, Gaussian mixture models, and computer vision to segment historical newspapers into article-level structure, reducing segmentation costs by 50% compared to manual processes.

Aug 2024 – May 2024 | NLP Algorithm Intern, Pachira Information Technology Co., Ltd. | Zhuhai, China
Supervisor: Dr. Chenglong Ma
- Implemented warning-light query handling, multi-intent recognition, and refusal logic for a RAG-based in-car voice assistant for Toyota vehicles.

Publications

C = Conference S = In Submission

[C.1] Beyond Readability Metrics: Plain Language Priorities in Disability Advocacy Organizations
Kumar, A., Glazko, K., **Sun, Y.**, Harniss, M., Wang, L. L., & Mankoff, J.
ACM FAccT 2026

[C.2] Viral Images: Identifying Reprintings within 1.5 Million Newspaper Photographs in Chronicling America
Buccalon, B., **Sun, Y.**, & Lee, B. C. G.
Accepted to CHR 2027 — Long Oral Presentation

[S.1] Easy Reading Is Not Deep Reading: A Classroom Study of AI-Assisted Elaborative Interrogation on Reader Comprehension
Kumar, A., **Sun, Y.**[*], Memon, S. A.[*], & Wang, L. L.
Under review, ACM CHI 2027 — [*]Equal contribution

Teaching Experience

Present – Sep 2026 | Reader/Grader, University of Washington Information School | Seattle, WA
IMT 521: Artificial Intelligence Foundations, *Instructor: Prof. Aylin Caliskan*

May 2025 – Jan 2025 | Reader/Grader, University of Michigan Department of Mathematics | Ann Arbor, MI
MATH 417: Matrix Algebra I, *Instructor: Prof. Ruiyuan Chen*

Dec 2023 – Sep 2023 | Undergraduate Tutor, University of Michigan Math Learning Center | Ann Arbor, MI
Calculus I–IV, *Supervisor: Irina Arakelian*

Leadership & Service

Present – Sep 2026 | Global Sustainability Fellow, Impending Bloom | Remote
Applied systems thinking and sustainability reporting frameworks to analyze recycling systems, identify data and reporting gaps, and develop evidence-based system models and recommendations.

Aug 2026 – Mar 2026 | Program Lead, Develop for Good | Remote
Oversaw four interdisciplinary student volunteer teams, coordinating project execution, client communication, and mentor engagement across nonprofit software development initiatives.

Feb 2026 – Oct 2025 | Product Manager, Develop for Good — PainUSA Team | Remote
Led five undergraduate student volunteers to prototype a web application for PainUSA, featuring an interactive clinician lookup map and chronic pain resources.

Skills

Research Methods	Mixed-Methods Evaluation, Experimental Design, Statistical Modeling, NLP, Computer Vision, Information Retrieval
Programming	Python, R, SQL, Bash, JavaScript/TypeScript, Go, C/C++
ML & Data	PyTorch, scikit-learn, Pandas, NumPy, OpenCV, LangChain
Systems & Platforms	Git, Docker, GCP, AWS, Terraform, FastAPI